Database Management Portal for Breast Cancer Screening

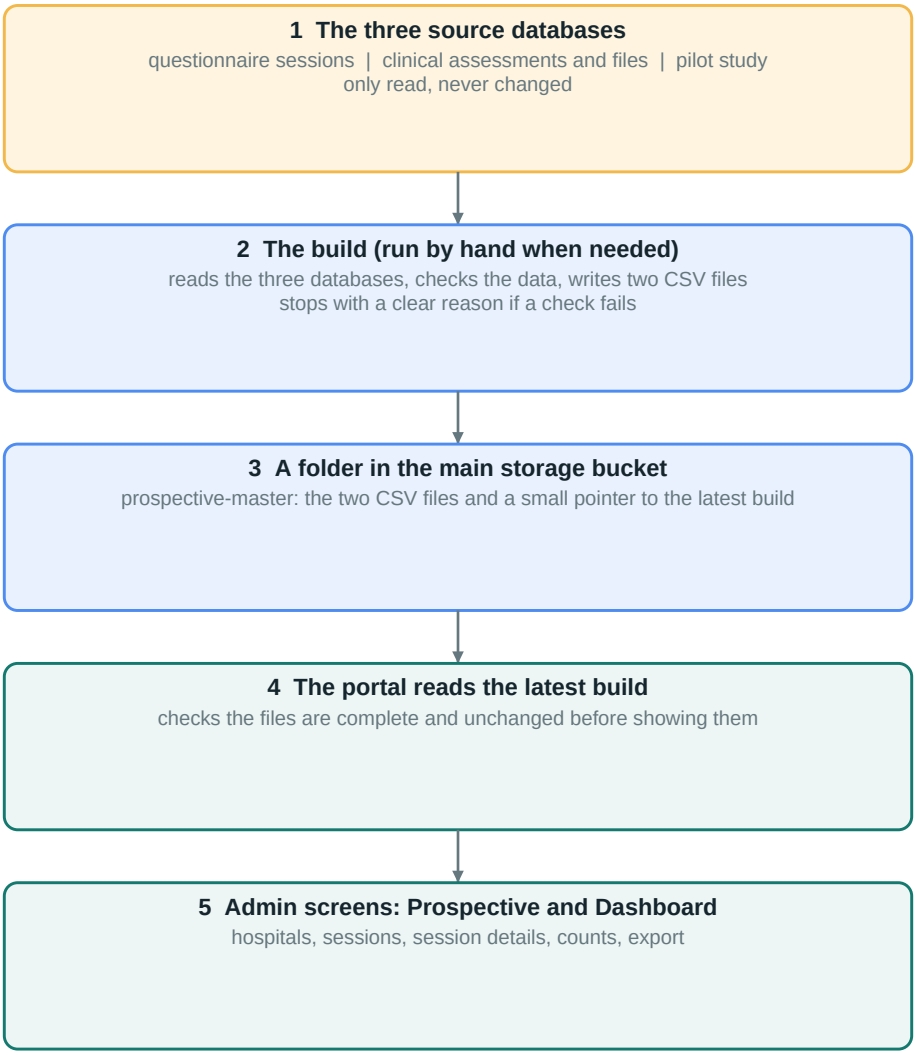
Prospective study and Dashboard

How the data gets into the portal, and what the admin can see

Who	What they can do
Admin	Opens the Prospective screens and the Dashboard. Filters sessions by hospital, CR / DR, BI-RADS, density, completeness and more. Opens one session to see its answers, assessment and files. Exports a list of file paths or a filtered CSV.
Mammo Tech and Radiologist	No access to Prospective or the Dashboard. These screens are for admins only.

Numbers in this document are from the first real build on 30 October 2026 and will change with each new build.

1. How the data gets in



The portal does not look into the live databases while you use it. A separate build collects the data into two CSV files, one row per questionnaire session and one row per file. The portal reads those files. This keeps the live clinical databases safe and makes every screen fast.

- Run the build when you want fresh numbers. It takes about a minute.
- Each build is saved as its own version, so an older one can be brought back if needed.
- The portal refuses a build that is incomplete or has been changed after it was written.

2. What is in the first build

Item	Count
Sessions included	3,466 (portal and pilot)
Files listed	12,749 (mammogram views, reports, ultrasound, annotations, consent photos)
Left out	1,225 sessions that were started but never filled in, and 6 pilot test records without a session
Hospitals	25 hospital records; each pilot site is shown under its parent hospital

3. The rules behind the numbers

- **CR or DR** of a hospital is taken from the hospital record in the database, as given by the doctors. Where a pilot site and its parent hospital differ, each keeps its own value.
- **Machines** are shown as recorded: how many, how many different ones, their names, makes and technology.
- **Test data** is hidden by default and not counted: the Test hospital, Tanuh Foundation, the Pilot Study test site and the BISICON Conference pilot. A tick box shows it when needed.
- **Right and left breast**: in the database the mammogram columns hold the right breast and the ultrasound/biopsy columns hold the left. The build corrects this, so the portal always shows right and left correctly.
- **A complete datapoint** means: consent (not needed for pilot), the questionnaire, all four mammogram views, and BI-RADS and density for both breasts.
- **One woman can have several sessions.** A tick box shows only her latest session.
- **Privacy**: hospital contact names, emails and addresses are not copied into the files. Subject IDs are visible to admins only.

4. The screens

Prospective

- **Summary cards**: sessions, complete datapoints, sessions with a clinician assessment, and CR / DR counts. They follow the filters.
- **Hospitals**: one card per hospital with its CR or DR, machines, and counts of sessions, complete and pilot. Click a card to filter to that hospital and its pilot sites.
- **Sessions table**: search, Portal or Pilot, CR or DR, BI-RADS, density, complete only, latest per woman, include test data. Click a row to open it.
- **Session details**: overview, questionnaire answers and risk results, the clinician assessment, what is complete or missing, and the list of files.
- **Export**: a list of file paths for the filtered sessions, or the filtered table as a CSV.

Dashboard

- **Prospective**: sessions by month, CR vs DR, how complete the data is, BI-RADS and density, and sessions by hospital.
- **Retrospective**: cases received, segmented, in progress, rejected, where the cases are, and who supplied them. These numbers are live.
- The Dashboard updates itself every minute while it is open.

5. Running and updating it

Task	How
See the latest numbers	Open Prospective or Dashboard as an admin.
Refresh the data	Run the build again (from the Cloud Shell, about 15 minutes with the setup steps) and publish it to the same folder. The portal picks up the newest build within a minute.
Change a hospital's CR or DR	Change it on the hospital record in the database, then rebuild.
Mark another site as test data	Add it to backend/src/data/test_hospitals.csv and deploy.
Go back to an older build	Point the pointer file (current.json) at the older build's folder.

6. What the build reads

Read only: SELECT queries inside a read-only transaction, on a fixed list of columns. Contact names, emails, addresses, IP addresses and passwords are never read.

Database	Tables used	What for
bcd_questionnaire	session_table, session_data_table	Session start and end, consent photo, portal answers, Snehitha risk
bcd_application2	hospitals, machines, doctor_assessments, attachments, users, roles	Hospital and CR / DR, machines, BI-RADS and density, findings, the list of files
pilot_deployment	patient_details, pilot_questionnaire, pilot_result, subjects	Pilot details and answers, pilot risk result, linking a woman's portal and pilot sessions

Checks before anything is published

- Every session appears once; every file belongs to a session in the table; every file address is in the main bucket.
- BI-RADS only 0 to 6 and density only A to D; times carry the Indian time offset; yes/no columns are only 0 or 1.
- Each session has exactly one latest file of each type; the summary columns (all four views, report, ultrasound, biopsy) agree with the file list.
- A complete datapoint really has all four views, BI-RADS and density for both breasts, and consent (not needed for pilot).
- Unknown file types, or pilot rows without a session, are reported. If any check fails, nothing is published and the reason is printed.

7. Running the build (Google Cloud Shell)

Needs your MySQL login (the one used in MySQL Workbench) and the Google account used for the console. Use two Cloud Shell tabs.

Tab	Command	What it does
Both, first time only	mkdir -p kit; cd kit; unzip -o ../prospective_build_kit.zip; pip install --user sqlalchemy pymysql cryptography google-cloud-storage	Unpack the build kit and install what it needs
Tab 1 (leave running)	./cloud-sql-proxy bcd-prototypes:asia-south1:tanuh-bcd-questionnaire-dev --port 3307	Opens a safe connection to the database with your Google login
Tab 2	./run_build.sh /tmp/prospective-built	First build into a local folder, to look at. Asks for the MySQL user and password (not saved)
Tab 2	./run_build.sh gs://breast-cancer-image-dataset/prospective-master	Publish: writes a new build folder and updates current.json

After publishing, delete the local copy (rm -rf /tmp/prospective-built; it holds subject IDs), stop the proxy with Ctrl+C, and close both tabs.

8. Connecting the portal (once)

The portal needs one setting that says where the latest build is. It is added to the server's settings file, then the backend is restarted. The app lives in /home/bharath_user/qc_portal on the VM qc-bcd-portal-instance.

```
echo 'PROSPECTIVE_MANIFEST_URI=gs://breast-cancer-image-dataset/prospective-master/current.json' |
sudo tee -a /home/bharath_user/qc_portal/.env
sudo docker compose -f /home/bharath_user/qc_portal/docker-compose.prod.yml --project-directory
/home/bharath_user/qc_portal up -d --force-recreate backend
```

Run the first line only once. After that, every new published build is picked up automatically within a minute.

9. When something goes wrong

Message or symptom	Meaning and what to do
Build failed (OperationalError)	The database login or connection failed. Check the MySQL user and password, and that Tab 1 still shows the proxy running.
Build failed (ValueError): ...	One of the build's checks found a data problem; the sentence says which. Send it to the developer. Example seen: pilot rows without a questionnaire session (6 test records), now skipped and reported.
Old message without a reason	The kit is an old version. Delete the zip in Cloud Shell, upload the new one, check its size, unpack again.
Prospective says the data source is not set up	The setting in section 8 is missing, or the backend was not restarted.
Prospective shows an error about checksum or counts	The build files changed after they were written, or a build is half written. Run the build again.
Hospital shows Not set instead of CR / DR	The hospital record has no CR or DR. Fill it in the database and rebuild.
Numbers look old	Nothing refreshes by itself yet: run and publish the build again.

10. What comes next

- Open the actual DICOM images and PDF reports from a session inside the portal.
- Decide how often the build is refreshed, and who runs it.
- Decide what management wants on the Dashboard, and whether subject IDs should be hidden for some admins.